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F-ViTA: Foundation Model Guided Visible-to-Infrared Translation

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Abstract

Thermal imaging is crucial for scene understanding, particularly in low-light and nighttime conditions. However, collecting large thermal datasets is costly and labor-intensive due to the specialized equipment required for infrared image capture. To address this challenge, researchers have explored visible-to-thermal image translation. Most existing methods rely on Generative Adversarial Networks (GANs) or Diffusion Models (DMs), treating the task as a style transfer problem. As a result, these approaches attempt to learn both the modality distribution shift and underlying physical principles from limited training data. In this paper, we propose F-ViTA, a novel style transfer approach that leverages the general world knowledge embedded in foundation models to guide the diffusion process for improved translation. Specifically, we condition an InstructPix2Pix Diffusion Model with zero-shot masks and labels from foundation models such as SAM and Grounded DINO. This allows the model to learn meaningful correlations between scene objects and their thermal signatures in infrared imagery. Extensive experiments on five public datasets demonstrate that F-ViTA outperforms state-of-the-art (SOTA) methods. Furthermore, our model generalizes well to out-of-distribution (OOD) scenarios and can generate Long-Wave Infrared (LWIR), Mid-Wave Infrared (MWIR), and Near-Infrared (NIR) translations from the same visible image. Code: Released post-review.

1. Introduction

Many critical applications in autonomous driving, robotics, and surveillance require a precise understanding of natural scenes that remains robust in low-light and nighttime conditions. Thermal imaging has been widely explored for scene understanding and object recognition in such environments, where external illumination is impractical due to various constraints.

The infrared spectrum is typically divided into two regions: a reflection-dominated region, which includes the Near-Infrared (NIR) and Shortwave Infrared (SWIR) bands,

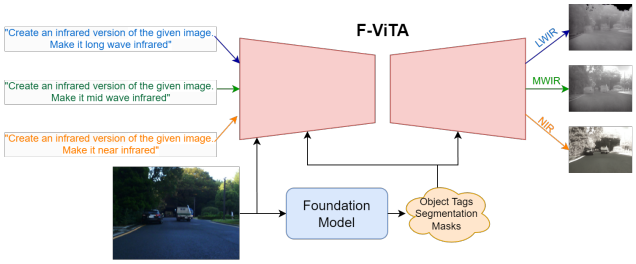


Figure 1. Our model, F-ViTA, leverages pretrained foundation models to extract object tags and segmentation masks from visible images in a zero-shot manner, using this information to enhance translation to the thermal domain. Additionally, F-ViTA enables user-guided infrared image generation through text prompts, allowing for the synthesis of specific infrared types–an ability not explored in existing methods.

and an emission-dominated thermal region, comprising the Midwave Infrared (MWIR) and Longwave Infrared (LWIR) bands. Given its resilience to challenging lighting conditions, infrared imagery provides a valuable source of information for these applications [3, 5, 8, 15, 22, 30].

Recent studies have explored RGB-IR fusion to enhance tasks such as pedestrian detection [7, 23], segmentation [28, 31], and object detection [18, 34]. These approaches typically rely on paired RGB and IR images from the same scene. However, capturing IR images requires specialized hardware, including infrared cameras and sensors, making large-scale dataset collection both costly and labor-intensive. As a result, IR datasets remain relatively scarce in the literature—especially paired RGB-thermal data—driving research efforts toward automatically generating infrared images from visible images.

Various methods have been proposed for image translation using Generative Adversarial Networks (GANs) [14, 17, 36] and Diffusion Models (DMs) [4, 20, 21]. These approaches aim to learn the distribution shift between the two domains while preserving the physical properties of infrared images. However, in the low-data regime of RGB-to-IR translation, such models are highly prone to overfitting, which significantly limits their generalizability.

Recently, foundation models (FMs) have been intro-

duced that are trained on large-scale natural data from the internet. These models learn highly representative features for visual data that are beneficial for various downstream tasks and promote generalized performance. Some examples include CLIP [24] for image-text alignment, Grounding DINO [19] for open-world object detection, Segment Anything (SAM) [13] for prompted image segmentation and SAM2 [25] for prompted video segmentation. Given the generalizable nature of such models, using the knowledge contained in them can help address the limited data issues with the current RGB-IR translation methods. Motivated by this, we propose **F-ViTA**, which uses implicit guidance from FMs to improve the translation quality. For a given natural image, we first pass it to the FM to extract labeled masks of all objects present in the scene. Then, we inject this information during the diffusion process during training the translation model, which is based on Instruct Pix2Pix [1]. This can encourage the model to correlate the output pixel intensity of the masked regions with the label of the object. For example, regions with the label "person" should be given a higher pixel intensity in the IR image. Such guidance also allows our model to do well on datasets with limited training pairs. Thus, our contributions are as follows:

1. We present F-ViTA, a Diffusion Model-based approach for visible-to-infrared image translation that utilizes outputs from pretrained FMs to guide the translation process.
2. We show that F-ViTA outperforms existing SOTA methods on five public datasets from NIR, MWIR and LWIR categories on multiple metrics.
3. We also show that F-ViTA is able to generalize well to OOD data and can also generate the translated IR image from any of the three bands (LWIR, MWIR, NIR) given the user intent through a text prompt, as shown in Fig. 1. This is one of the first methods that shows such a functionality for visible-to-IR translation.

2. Related Work

Visible-to-thermal translation has been widely studied in the literature, with a majority methods attempting to solve this problem using GANs. Some early methods use Pix2Pix [11], an image-to-image translation model which uses conditional adversarial networks and cycle-GAN [35], which adds an additional consistency loss for stable training. These methods are not specific to visible-to-thermal translation. ThermalGAN [14] was introduced for this task, with a focus on person identification using the thermal signature. InfraGAN [36] is also a GAN-based approach that performs translation to the infrared domain for general scenes using pixel-wise loss functions and structural similarity. Recently, EGGAN [17] was introduced that uses information from the edge map of the natural scene to guide the translation process. For this, they add an additional constraint to preserve the edges in the image.

As an alternative to the above GAN-based approaches, PID [20] was recently introduced, which uses DMs to perform the translation. In addition, this paper introduces physics-motivated loss functions that preserve the physical characteristics of the generated IR image. However, PID requires two-stage training, where the dataset-specific physics parameters are learnt first and then used as ground truths during the translation training. Furthermore, this work uses a common set of parameters for the dataset as a whole, rather than have a physical parameter for every object in the scene and has only been proposed for LWIR. Our method also uses a DM for translation, but it differs from PID in being a single-stage end-to-end approach. In addition, unlike PID, which explicitly learns dataset-level parameters, we implicitly try to learn object-level correlations by passing labeled maps during the diffusion process. Moreover, we show that our method produces good results for LWIR, MWIR as well as NIR images. TIRControlNet [21] also uses a diffusion model with segmentation masks as conditioning. However, this work expects precise segmentation masks as an additional input. In our case, we generate pseudo segmentation masks which are used as a weaker signal. In addition, TIR-ControlNet is not a RGB-to-IR method but instead it uses segmentation maps and text to generate new thermal images. In our case, we train the method to generate an IR image for the corresponding RGB image.

3. Methodology

3.1. Preliminaries: Grounded-SAM

Given a visible image, we use Grounded-SAM [26] to produce labeled masks. Grounded-SAM assembles Grounding-DINO [19], which is an open-world prompted object detector, and SAM [13], which is a promptable image-segmentor. Given a text prompt and an image, Grounding DINO can detect the object corresponding to the text input and shows a strong zero-shot performance thanks to its training data. Similarly, SAM was trained on 11 million images and 1.1 billion masks and exhibits a strong zero-shot performance for segmenting out an object corresponding to a point, box or mask prompt. Thus, given a text descriptor of an object, Grounded-SAM first employs Grounding DINO to extract a bounding box for that object and passes it to SAM as a prompt, thus generating the labeled mask, where the input text is the label.

RAM-Grounded-SAM (RGS): Grounding-SAM can be used for automatic dense image annotation [26], where given an RGB image, the output is a list of masks for the objects in the scene, which are annotated with their corresponding text labels. For this, a foundational image tagging model like Recognize-Anything (RAM) [33] is used. RAM

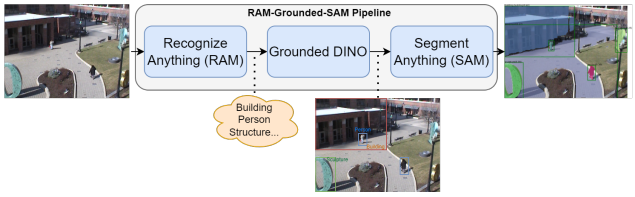


Figure 2. An example usage of the RAM-Grounded-SAM pipeline. The dotted lines indicate the output at every step.

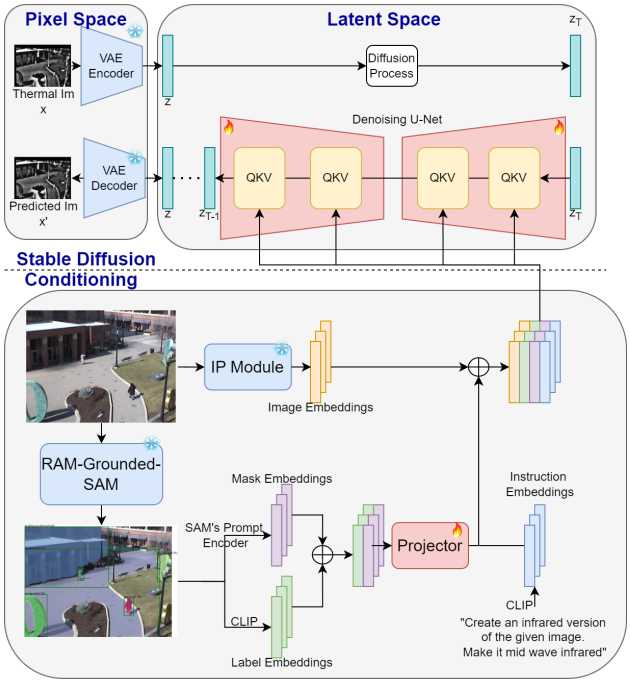


Figure 3. Training pipeline. The stable diffusion part is responsible for learning the distribution of thermal images. The conditioning part provides the visible image to enforce structural similarity as well as guidance from the foundation model and text instructions for improved translation. Only the Denoising UNet and the Projector module are trainable in the pipeline.

is trained to detect all objects in a scene and generate tags for them. It is trained on a large corpus of image-caption pairs from the internet, which are parsed to obtain the image with its corresponding tags. Given these pretrained foundation models, the RAM-Grounded-SAM pipeline passes an image through the RAM model to generate tags for all the present objects. These tags are passed to the Grounding-DINO model to generate corresponding bounding boxes, which are further passed to SAM to generate masks. Note that this entire pipeline does not involve any trainable parameters. Fig. 2 shows an example of the pipeline output. These foundation models are pretrained on a vast pool of diverse datapoints and thus are able to produce valuable priors for tasks like visible-to-infrared translation that are based in a low-data regime. More discussion in supplementary.

3.2. Model Architecture

We use Stable Diffusion [27] for the translation process, similar to Instruct-Pix2Pix [1]. As shown in Fig. 3, we use a pretrained encoder to convert the thermal image to the latent space, where it undergoes the forward diffusion process and the backward denoising process. The denoising is done using a UNet conditioned with additional prompts to control the generation process. The conditioning involves attention layers for each U-Net block, where the keys and the values are generated using the visible image and the text instructions and the queries are latent vectors at different timesteps. Finally, the pretrained decoder converts the generated latent back to the pixel space. In Instruct-Pix2Pix, the conditioning described before is done using an original image and a detailed text instruction, which describes the user-intended change to the original image. In contrast, F-ViTA also adds two additional conditioning embeddings derived from foundation model outputs.

We pass the visible image through the frozen RAM-Grounded-SAM pipeline, which generates a number of text-annotated masks for the objects present in the image. We pass these masks through the prompt-encoder provided by SAM to generate discriminative embeddings. In addition, we pass the text-labels for every object through the pretrained CLIP encoder to generate text embeddings. These are concatenated along feature axis and passed through a trainable projector. This module is responsible for learning a transform that converts the FM embeddings to the space expected by Stable Diffusion. These are then concatenated to the image embeddings, obtained by passing the visible image through Instruct-Pix2Pix's IP module, and instruction embeddings, which are CLIP embeddings of the text instruction. The latter two embeddings are unique for a given image-instruction pair, while the former two vary for every object. Hence, we repeat the latter two embeddings to be equal to the number of detected objects, for concatenation. For the text instruction, we generate 50 synonymous sentences for the text: "Create an infrared version of the given image. Make it $< type >$ wave infrared." Here, $< type >$ can take the values in $\{\text{near}, \text{mid}, \text{long}\}$. Thus, we generate a conditioning tensor using the image, information from the foundation models, and a text instruction.

We train the entire pipeline on visible-thermal image pairs. Only the projector module and the denoising U-Net are the trainable modules in the pipeline. Since the conditioning tensors contain information about an object and its location in the image, we expect the model to learn physical correlations between the heat signatures of the object and the pixel intensity in the generated thermal image at the location of the object, for example, generating higher intensity pixels at the location of the object "person". While existing methods expect the model to learn everything about the translation from the data, we utilize the guidance pro-

Method	FID (↓)	LPIPS (↓)	SSIM (↑)	PSNR (↑)
Pix2Pix [11]	199.06	0.38	0.30	16.34
ThermalGAN [14]	407.01	0.60	0.15	11.59
InfraGAN [36]	140.01	0.50	0.41	16.95
EGGAN-M [17]	139.11	0.43	0.29	10.32
EGGAN-U [17]	140.92	0.55	0.37	14.3
PID [20]	**84.26**	0.36	0.40	17.26
F-ViTA (Ours)	133.30	**0.23**	**0.73**	**18.73**

Table 1. Results on FLIR-ADAS [9] dataset. Bold - best, underline - second best

Method	FID (↓)	LPIPS (↓)	SSIM (↑)	PSNR (↑)
Pix2Pix [11]	132.04	0.22	0.69	21.25
ThermalGAN [14]	277.85	0.24	0.66	19.74
InfraGAN [36]	222.96	0.159	0.76	22.97
EGGAN-M [17]	79.45	0.37	0.48	10.65
EGGAN-U [17]	76.27	0.27	0.63	17.07
PID [20]	**51.69**	**0.14**	0.79	**23.6**
F-ViTA (Ours)	76.18	0.21	**0.88**	18.96

Table 2. Results on KAIST dataset [10]. Bold - best, underline - second best

Method	FID (↓)	LPIPS (↓)	SSIM (↑)	PSNR (↑)
EGGAN-M [17]	542.51	0.47	0.55	12.95
EGGAN-U [17]	75.76	0.18	**0.86**	20.46
PID [20]	85.58	0.25	0.58	14.1
F-ViTA (Ours)	**61.97**	**0.15**	0.78	**20.92**

Table 3. Results on the OSU dataset [6]. Bold - best, underline - second best. Our implementation of InfraGAN, Pix2Pix and ThermalGAN suffered from mode collapse and almost always produced random outputs. This may be due to the low size and variability of the dataset. Hence, we did not add them to the results.

Method	FID (↓)	LPIPS (↓)	SSIM (↑)	PSNR (↑)
InfraGAN [36]	241.42	0.36	0.87	17.15
EGGAN-M [17]	238.41	0.35	0.93	12.81
EGGAN-U [17]	152.09	0.23	0.95	18.45
PID [20]	179.65	0.20	0.95	19.22
F-ViTA (Ours)	**118.52**	**0.16**	**0.96**	**27.55**

Table 4. Results on the LiTiV2012 dataset [16]. Bold - best, underline - second best. Our implementation of InfraGAN, Pix2Pix and ThermalGAN suffered from mode collapse and almost always produced random outputs, possibly due to the low size and variability of the dataset. Hence, we did not add them to the results.

vided by FMs to reduce the load by providing localization and object information. Hence, our method is able to outperform existing methods in low-data regimes.

4. Experiments and Results

4.1. Image Translation Experiments

We compare our method with existing SOTA visible-to-thermal translators on five datasets, namely KAIST [10], FLIR-ADAS [9] (LWIR), OSU [6], LiTiV [16] (MWIR) and NIRSCENE [2] (NIR). Details about the datasets and the implementation details are in supplementary. The results on FLIR-ADAS are tabulated in Tab. 1. We find that our method is able to outperform existing methods on multiple metrics, with a significant rise observed in SSIM. However, we see a lower FID score than the current SOTA. While FID compares the distributions between the ground truth and the generated thermal image, SSIM measures the structural and visual soundness of the generated image. Hence, both are important to gauge the quality of generation. On the KAIST dataset, as can be seen in Tab. 2, we see a similar trend where the SSIM for our method is better but other metrics are comparable to PID. This is not surprising, given that PID models the physical quantities behind LWIR images and constrains the model accordingly. In contrast, our model captures the structure of objects better due to localization information present through the masks (causing improved SSIM), but at the same time has a slight tendency to miss the differences in output intensities for different parts of the same object (causing increased FID). This can happen since there are multiple similar signals from the prompts for different parts of the same object. In addition, note that FLIR is a smaller dataset than KAIST, which is why we see an overall better performance with our method.

Method	FID (↓)	LPIPS (↓)	SSIM (↑)	PSNR (↑)
EGGAN-M [17]	126.2	0.18	0.83	15.25
EGGAN-U [17]	125.6	0.23	0.79	18.45
PID [20]	166.67	0.31	0.62	14.21
F-ViTA (Ours)	**63.67**	**0.11**	**0.85**	**19.23**

Table 5. Results on the NIRScene dataset [2]. Bold - best, underline - second best

Tab. 3 shows the results on the OSU dataset, where we see a significant improvement in all metrics, while SSIM is comparable. Similarly, for LiTiV, as shown in Tab. 4, and for NIRScene, as seen in Tab. 5, we find that our method outperforms all other methods across all metrics. Note that these are cases of low-data regimes, where the advantages of our model are clearly demonstrated. We present qualitative results in Fig. 4 across all five datasets used for comparison, selecting examples from both indoor and outdoor scenes. As shown in the figure, our method (third column) performs consistently well across all datasets. PID excels in LWIR but struggles to generate accurate translations for MWIR and NIR. This is expected, as the theoretical analysis and formulation of PID are primarily designed for LWIR translations. In contrast, EGGAN-M and EGGAN-U slightly distort object structures, as observed in the figure. In comparison, our method, F-ViTA, generates thermal images that closely resemble the ground truth.

F-ViTA, being a Diffusion Model (DM), takes more time

Method	FID (↓)	LPIPS (↓)	SSIM (↑)	PSNR (↑)
PID [20] (tuned on KAIST)	196.10	0.39	0.85	13.53
Ours (tuned on KAIST)	**70.98**	**0.30**	**0.87**	**16.33**
PID [20] (tuned on FLIR-ADAS)	142.60	0.47	0.70	12.02
Ours (tuned on FLIR-ADAS)	**100.54**	**0.35**	**0.81**	**12.85**
Ours (tuned on all datasets from all wavelengths)	94.5	0.37	0.83	13.72

Table 6. OOD Results on the MFNet dataset [29]. Bold - best, underline - second best

than its GAN counterparts. For eg. on the LiTiV dataset, F-ViTA takes 3.25 s out of which 0.2s is due to RGS. In contrast, InfraGAN takes 0.14 s and PID takes 11.6s. Still, we believe that this shortcoming is covered by F-ViTA's metrics and applicability. More discussion in the supplementary.

4.2. Out-of-Distribution (OOD) Experiments

We use the MFNet [29] dataset to evaluate the generalizability of our method on out-of-distribution (OOD) data. MFNet contains visible images and their thermal counterparts in the LWIR range. As shown in Tab. 6, F-ViTA, trained on the KAIST dataset, performs significantly better on MFNet compared to PID trained on KAIST. Similarly, for models tuned using only the FLIR dataset, our method outperforms PID trained on the same data. PID first learns estimates for physical quantities related to the thermal dataset as a whole, while our method implicitly learns object-wise estimates. As a result, our approach exhibits greater robustness when transferred to a new dataset, as reflected in the table. Additionally, we conduct an experiment where we train F-ViTA using one dataset from each modality (KAIST for LWIR, LiTiV for MWIR, and NIRScene or NIR for NIR) and then test its performance on MFNet with the instruction to convert to long-wave infrared. We find that even with the inclusion of additional modalities, the performance on MFNet remains comparable to when trained using only the LWIR KAIST dataset.

4.3. Text-Prompted Translation Experiments

F-ViTA uses text instructions for guiding the diffusion model to perform the image translation. Hence, it is possible for it to learn to generate images for specific infrared spectra by specifying it in the instructions. To explore this, we train F-ViTA with the combined train set consisting of KAIST (LWIR), OSU (MWIR), LiTiV (MWIR) and NIRSCENE (NIR). We specify the type of IR it should generate using the text prompt. For instance, for KAIST, the text prompt reads, "Generate a LWIR image". Once trained, we test the model's ability to generate all three types of IR images from a given RGB image from the test data. Some examples of such translations can be seen in Fig. 5. From the figure, we observe that F-ViTA produces reasonable predictions for LWIR, MWIR, and NIR given a single RGB image. The GT column represents the ground truth thermal image from the dataset-specific wavelength range. The first five rows display images from the test sets of the datasets on which the model was trained. The last three rows show predictions on MFNet, which was not used during training, demonstrating strong out-of-distribution (OOD) performance with F-ViTA. The final row illustrates a failure case where the model misses pedestrians in the MWIR and NIR predictions, indicating potential areas for improvement. These failure cases are generally caused only when both the RGS and the DM miss the visual cues from the RGB image. However, it is fairly robust to misses only by RGS. More discussion in the supplementary. Notably, no existing methods demonstrate such multi-spectral translation capability. We are the first to explore this direction with F-ViTA and encourage further research in this area.

% original data used	MIoU (↑)	Mean Pixel Accuracy (↑)
0	0.501	55.58
10	0.503	56.01
25	0.520	59.22
50	0.526	59.1
100	0.527	59.6

Table 7. Results on MFNet [29] using Sigma [31] as the RGB-T segmentation pipeline. The rows indicate the percentage of real thermal data used in the training data, with the rest of the data being synthetic. We see only a small drop in the metrics as we go up the table (more synthetic data used), towards the applicability of our method. In comparison, using PID gives an MIoU of 0.491 and Mean Pixel Accuracy (MPA) of 54.64% at 0% and MIoU of 0.520 and MPA of 58.89% at 50% of data, showing effectiveness of our method over PID.

4.4. Downstream Applications

RGB-T Fusion for Segmentation: Thermal imagery is often used for multi-modal segmentation, where the RGB image and its paired thermal image are processed and fused to produce the output segmentation map. We conduct experiments to show an application of our pipeline for thermal image generation. We use the F-VITA model, tuned on the KAIST dataset, to generate thermal images for the MFNet [29] dataset. Next, we generate five types of data for training a RGB-T segmentation model, using 0%, 10%, 25%, 50%, and 100% of the actual thermal images respectively. The rest of the images are taken from the generated predictions from our model. We use the original thermal images for the testing data. We use the SOTA RGB-T fusion model, called Sigma [31] for the segmentation task. Sigma uses the recently introduced V-Mamba framework along with a concatenation and cross-attention mechansim to effectively fuse the two modalities. With this setup, we train Sigma with the five datasets and test its performance (Tab. 7). We observe that when only the original thermal data is used, Sigma gives a MIoU of 52.7 and mean pixel

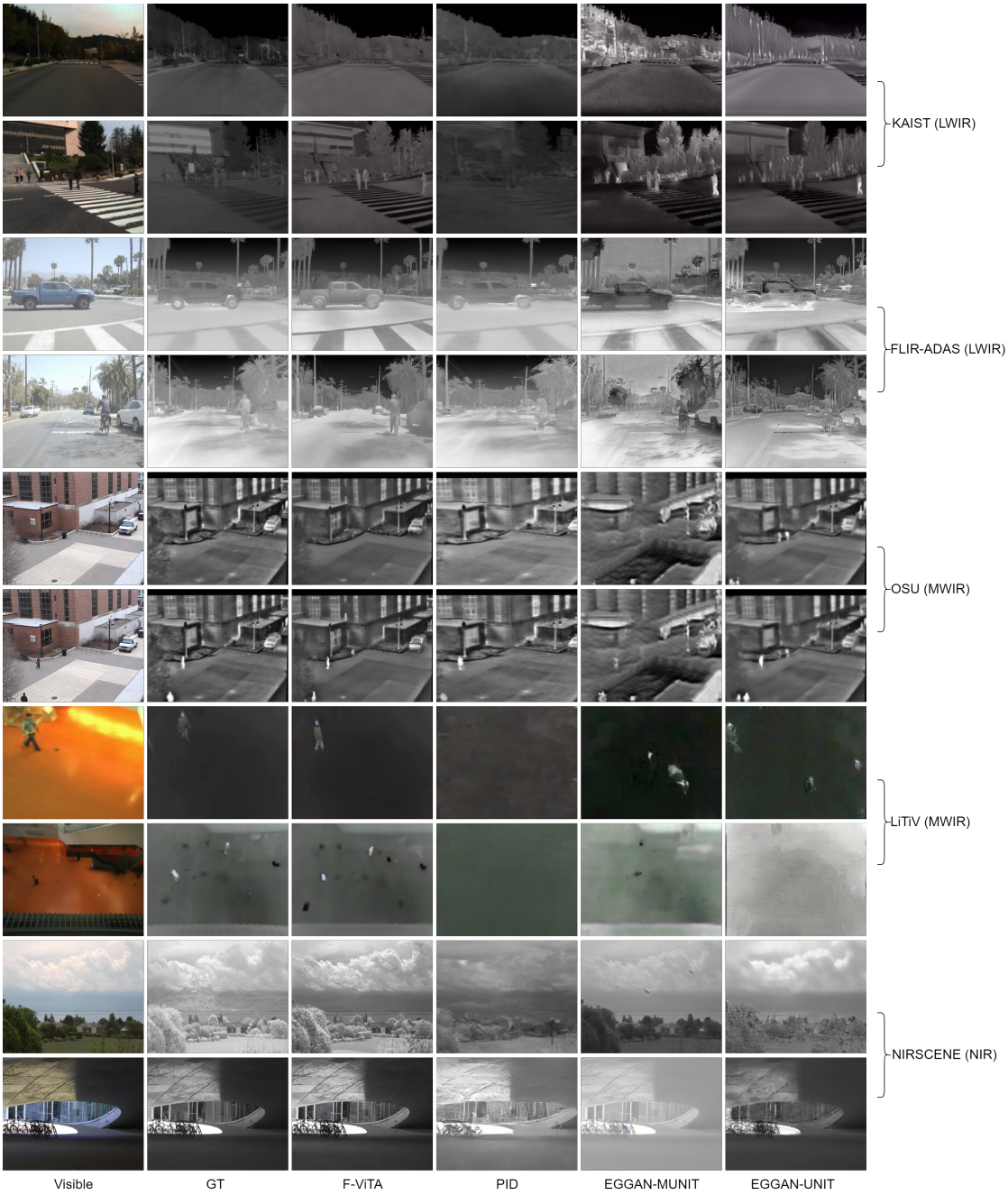


Figure 4. Qualitative Results over datasets from three different wavelength spectra. Our method (third column) is able to generate images more similar to the ground truth as compared to existing methods (fourth, fifth and sixth columns)

accuracy of 59.6. When we replace 50% of the data with the synthetic data generated with F-ViTA, the MIoU drops to 52.6 and accuracy to 59.1, which is not a significant drop compared to the reduction in original training data. With an all-synthetic training set, these numbers only reduce by 4 points. Note that the thermal data was created using F-ViTA trained on KAIST only, thus being OOD for the generation model. Thus, these results denote the quality of our predictions.

Thermal Image Segmentation: In this experiment, we train a Segformer [32] on the thermal training data using the same mixtures as in the previous experiment, and then evaluate its performance on the test data containing original thermal images. Note that we do not use RGB images here, making this a purely semantic segmentation task based on thermal images. As shown in Tab. 8, with 100% original

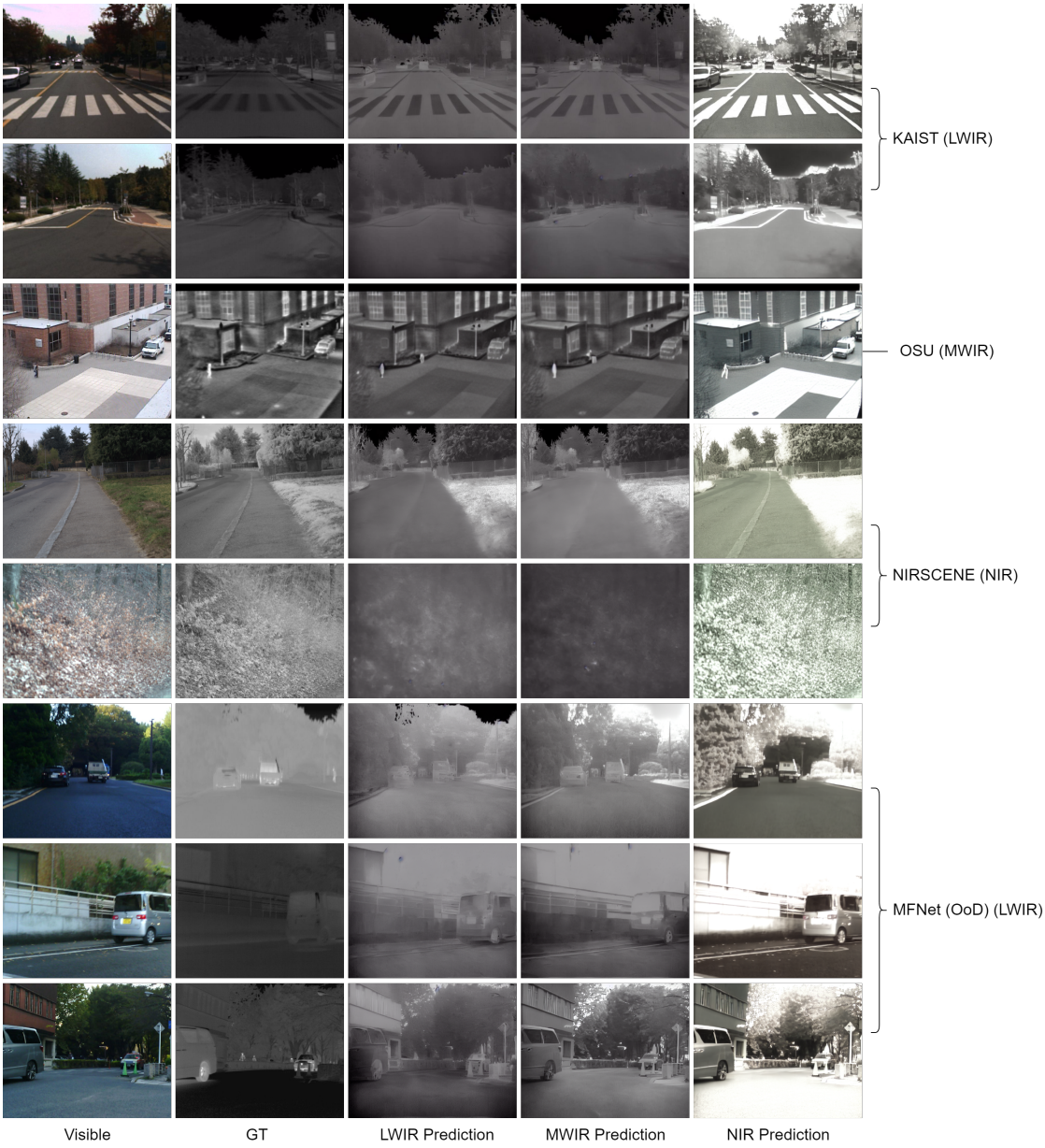


Figure 5. Text prompted translation capability of F-ViTA. Our method is able to generate LWIR, MWIR or NIR images based on the text instruction. The second column shows the ground truth which is from the wavelength range specified next to the dataset name on the right.

training data, we achieve a MIoU of 0.38 and a mean accuracy of 54%. Even when replacing 50% of the data with synthetic images, we observe comparable performance. Below this threshold, we see a noticeable decrease in accuracy, suggesting areas for further exploration. Nevertheless, the predictions from our method offer a viable alternative to capturing infrared images using specialized hardware.

Pedestrian Detection in Thermal Imagery: Pedestrian detection is an important application since it might not be easy to detect people from RGB images due to various lighting conditions (some examples in the supplementary). In this experiment, we use the thermal images generated by F-ViTA for the KAIST test dataset as synthetic data to generate five splits, with 0%, 10%, 25%, 50% and 100% being the real data percentage and the rest filled using synthetic data as done previously. Then, we train YOLOv8 [12] on these splits and test the performance on real thermal data, which is not used during training. As shown in Tab. 9, we calculate the precision (P), recall (R) and the mean average precision at 0.5 threshold (mAP@50) for all cases. We observe that even after replacing 50% of the real data, there is on par performance and surprisingly, a higher recall. As we

% original data used	MIoU (↑)	Mean Pixel Accuracy (↑)
0	0.30	43.38
10	0.30	45.24
25	0.30	40.01
50	0.38	51.29
100	0.38	54.12

Table 8. Results on MFNet [29] using Segformer [32] for thermal segmentation for 10 classes. The rows indicate the percentage of real thermal data used in the training data, with the rest of the data generated from F-ViTA. We see only a slight drop in the metrics as we go up the table (more synthetic data used). In comparison, using PID instead of our method gave MIoU of 0.23 and MPA of 21% at 0% original data and MIoU of 0.26 and MPA of 37.79% at 50% of original data used. This clearly shows the effectiveness of our method over PID.

% original data used	P (↑)	R (↑)	MAP@50 (↑)
0	0.63	0.45	0.48
10	0.64	0.56	0.57
25	0.64	0.60	0.60
50	0.67	0.62	0.62
100	0.67	0.59	0.62

Table 9. Results on KAIST [10] using YOLOv8 [12] for pedestrian detection. The rows indicate the percentage of real thermal data used in the training data, with the rest of the data being synthetic. We see only a slight drop in the metrics as we go up the table (more synthetic data used), which suggests a use case of F-ViTA for synthetic data generation. In contrast, using PID instead of F-ViTA gives 0.30 P, 0.15 R, 0.17 MAP at 0% original data, and 0.63 P, 0.59 R and 0.59 MAP at 50% of original data used. This shows the effectiveness of our method.

go up the table, we see that the drop in metrics is quite small compared to the amount of real data replaced, thus signifying the usefulness of the generated thermal images for such downstream tasks. Similar to the segmentation task, while we see that completely using the thermal data still has room for improvement, F-ViTA offers a viable solution for semi-supervised methods as seen from the table.

4.5. Ablation Studies

We perform an ablation study over the components extracted from the foundation models that are passed to the diffusion process. As seen in Tab. 10, when neither of text, bounding boxes or mask embeddings are used (first row), we see a lower performance in all metrics. Adding just the text tags identified from the image (second row) improves these metrics. The text tags are extracted using the Recognize Anything Model(RAM), and converted into embeddings using CLIP before passing to the DM. Next, we add only the embeddings from the bounding boxes for the objects detected by Grounding DINO (third row) and see on par metric values. However, adding mask embeddings from Grounded Segment Anything Model (Grounded SAM) (fourth row) gives us a slight improvement. This is expected since masks provide more precise localization than bounding boxes. In addition, unlike masks, a bounding box around bigger objects like buildings or trees may also encompass other objects like vehicles or people. Hence, we see an advantage when using masks. In the fifth row, we use both the text tags and the bounding boxes, which slightly improves the performance over second and third rows, where they are passed separately. Finally, the last row denotes the case used in F-ViTA, with both the text embeddings and mask embeddings being provided to the DM. We see that this case improves over all other rows, since text and masks provide the localization and semantic information about the objects to the DM. We do not include cases where box and mask embeddings are used together, since adding boxes is superfluous in the presence of masks.

Text Labels	Bounding Boxes	Masks	FID (↓)	LPIPS (↓)	SSIM (↑)	PSNR (↑)
			80.03	0.18	0.72	19.64
✓			76.43	0.20	0.74	20.5
	✓		76.01	0.20	0.75	20.01
		✓	69.81	0.19	0.76	20.7
✓	✓		75.18	0.17	0.76	20.24
✓		✓	**61.97**	**0.15**	**0.78**	**20.92**

Table 10. Ablation over the components passed to the diffusion process using the foundation model (FM). ✓indicates the component being used, with the first row indicating neither text, boxes or masks being used and the last row indicating F-ViTA where text and masks are used.

5. Conclusion

In this work, we introduce F-ViTA, a diffusion model-based visible-to-thermal image translator that leverages pretrained foundation models to guide the diffusion process. We demonstrate that incorporating object identification, location, and semantic information via segmentation masks and text labels significantly enhances translation, particularly in low-data regimes. Extensive experimentation on five public datasets across LWIR, MWIR, and NIR spectra shows that F-ViTA outperforms or matches existing methods on multiple metrics, with improved generalization on out-of-distribution datasets. Notably, F-ViTA is the first method to enable conversion of the same RGB image into LWIR, MWIR, or NIR based on user-provided text instructions, opening new avenues for future research. We also highlight the potential applications of visible-to-thermal translation in downstream tasks like RGB-T fusion and thermal segmentation, demonstrating its value for semi-supervised learning in data-limited scenarios. Future work should further explore text-prompted inter-modality transfer, with applications in biometrics, robotics, and surveillance.

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